Zero Waste Greenhouse

Nomenclature

Q Convective Heat Loss
R Thermal Resistance
K Thermal Resistivity (W/m².K)
GHG Greenhouse Gas Emissions
CO2e Carbon Dioxide Equivalent
GWP Global Warming Potential
A Area (m²)
TSI Solar Irradiance Wm⁻²
T Temperature (°C)
t Time (Seconds)
Kg Kilogram

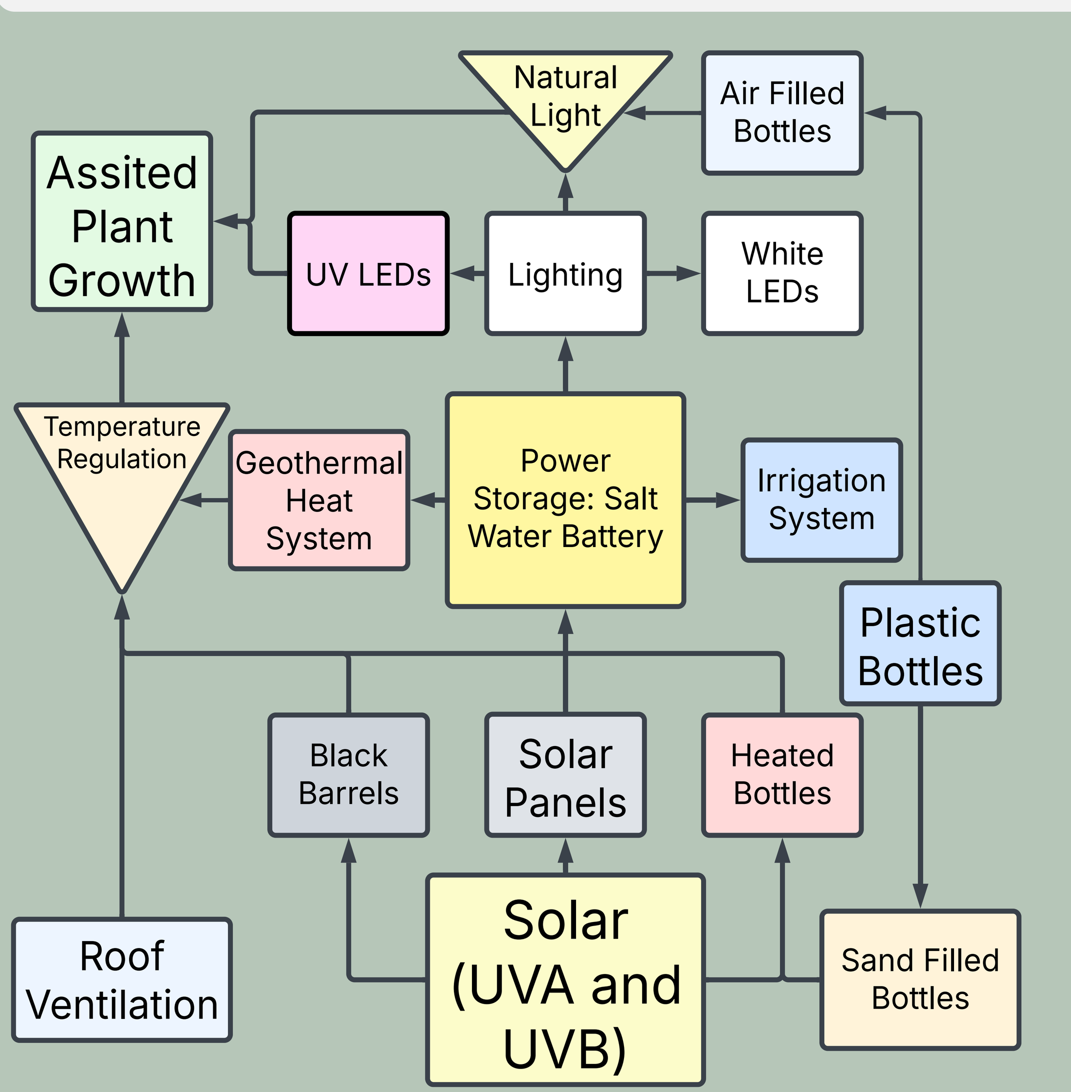
Introduction

Importance of Greenhouses:
Provide a controlled environment for crop growth.
Extend the growing season.
Help to maintain crop yield.

Problems in Making Greenhouses:
High energy consumption, heating and cooling.
High environmental impact from high energy usage.

Our Approach:
Sustainable focused.
Integration of energy-saving technologies.
Eco-friendly construction materials.
Efficient crop production with minimal environmental harm.

Process Flowchart



Cost Analysis

Material	Cost
Solar panel	400 watt solar panel £225, ≈ £2,800 (Heatable, 2025)
Sand	≈ £0.30 per kg (wholesale), £240 (Greenfingers, n.d.)
Geothermal pump	£12,000 6 kW system (IMS Heat Pumps, 2024)
Bamboo	≈ £6 (10mm x 5m), £11,460 (UK Bamboo, n.d.)
LED strips	£241.50 (100m), £724.50 (LED Supply and Fit, n.d.)
Crude tall oil	£0.52/kg on average, £130 (Business Analytiq, n.d.)
Salt battery	≈ £770/kWh, £7700 (McInerney, 2024)
	Total estimated cost = £35,054.50

We are initiating a community-driven program aimed at collecting used 2L plastic bottles and repurposing them into the construction of the structure.

Old polystyrene can be shredded, melted, or chemically broken down for reuse in products like insulation to minimizing environmental impact.

Large industrial drums/barrels can be sourced and then thoroughly cleaned and sealed for water collection or salt battery use.

Background & Rationale

Climate of Southeast England
The table below shows the conditions of summer and winter weather in Southeast England (Medway, Kent), using data collected from the MET Office, (2019) and considerations used for making a sustainable greenhouse.

Parameter	Summer	Winter
Average Temperature	Up to 30°C	Around -2°C
Daylight Duration	Up to 16 hours	Around 8 hours
Rainfall	Moderate	Moderate
Sunlight Availability	High, suitable for natural lighting	Low, needs artificial lighting and heating
Wind Speed	15–20 km/h (Increase during storms)	15–20 km/h (Increase during storms)
Climate Challenges	Overheating, ventilation needs	Heating demand, limited sunlight
Greenhouse Consideration	Ventilation & shading systems	Efficient insulation, lighting & heating

Why Sustainable Agriculture Matters Here
· Southeast England is a key agricultural region in the UK which produces a wide range of crops including fruits, vegetables and ornamental plants.
· Greenhouses offer a way to grow crops whole year, and it protect them from extreme weather. However, traditional greenhouses are energy-intensive which relies heavily on fossil fuels for heating and cooling (*GOV.UK, 2019).*
· Transitioning to sustainable greenhouse practices is essential to reduce greenhouse gas emissions, conserve energy and support local food production.

Environmental Challenges in the Region
· With industrial activity there is potential environmental contamination which can be counteracted with carbon sequestration.
· Redeveloping for agricultural purposes requires careful planning to ensure sustainability and avoid long-term environmental damage *(Environment Agency, 2021).*

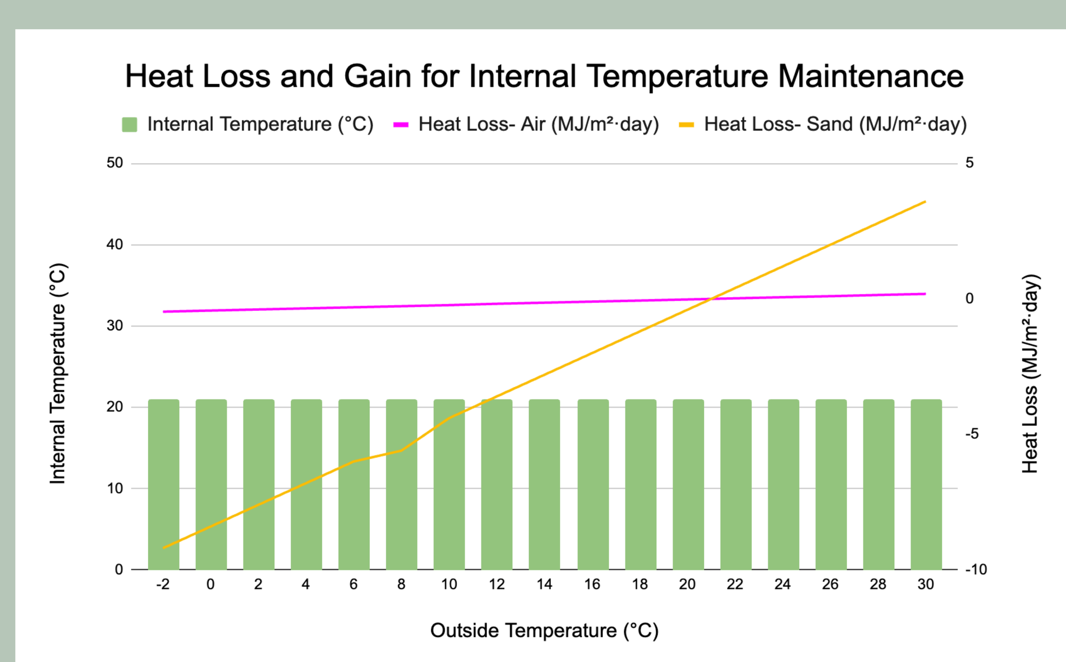
Thermal Performance

Material	k (W/m².K)	l (m)	R Total (K/W)
Air	0.026	0.108	4.153
Sand	0.7	0.108	0.154
Plastic	0.2	0.001	0.0005

Fourier's Law of Heat Conduction:

$$R_{Total} = \left[\frac{l}{k}\right]$$

$$\dot{Q} = \left[\frac{A(T_1 - T_2)}{R_{Total}} \times t\right]$$



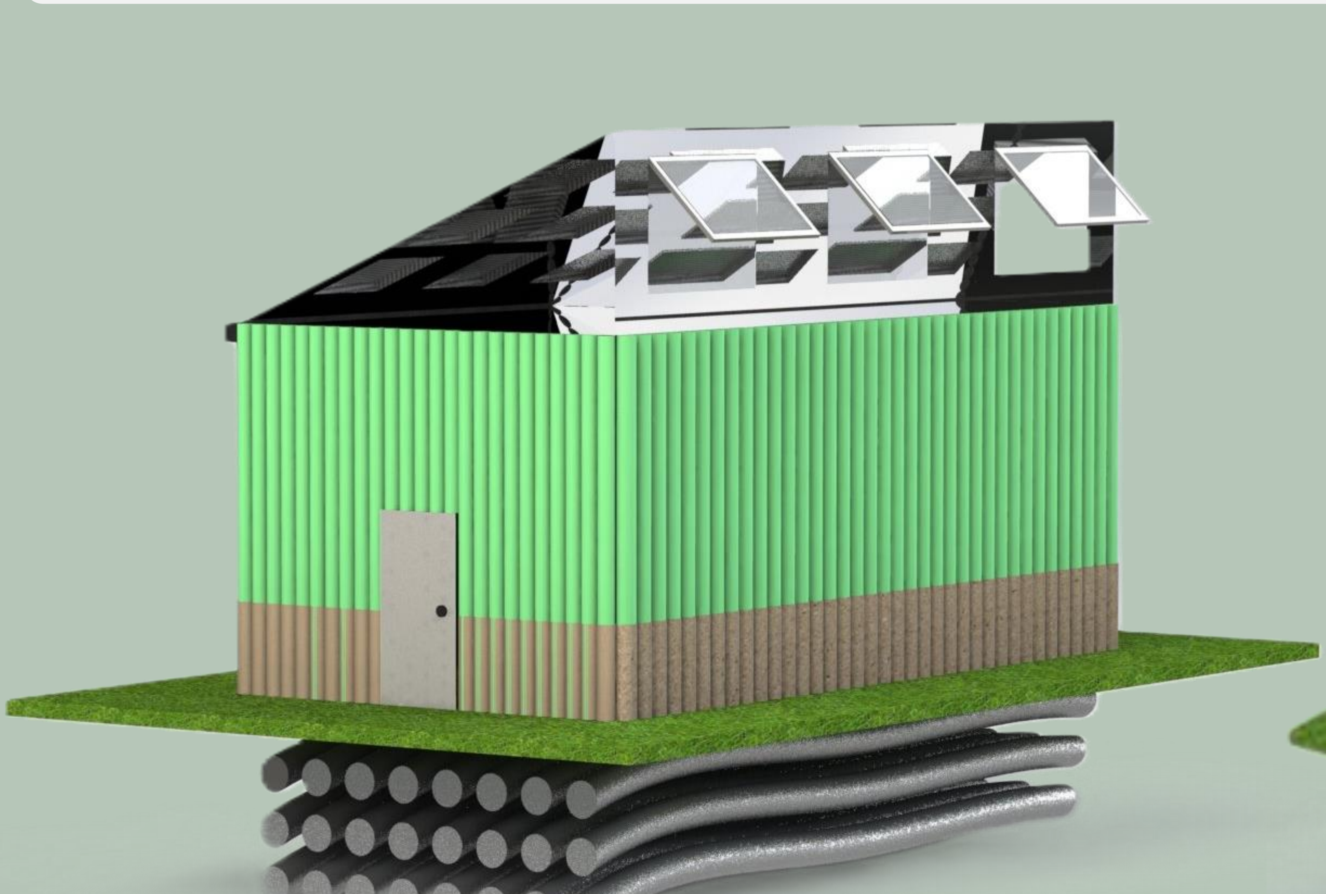
TSI ≈ 600 W/m² with 80% transmission through PET:

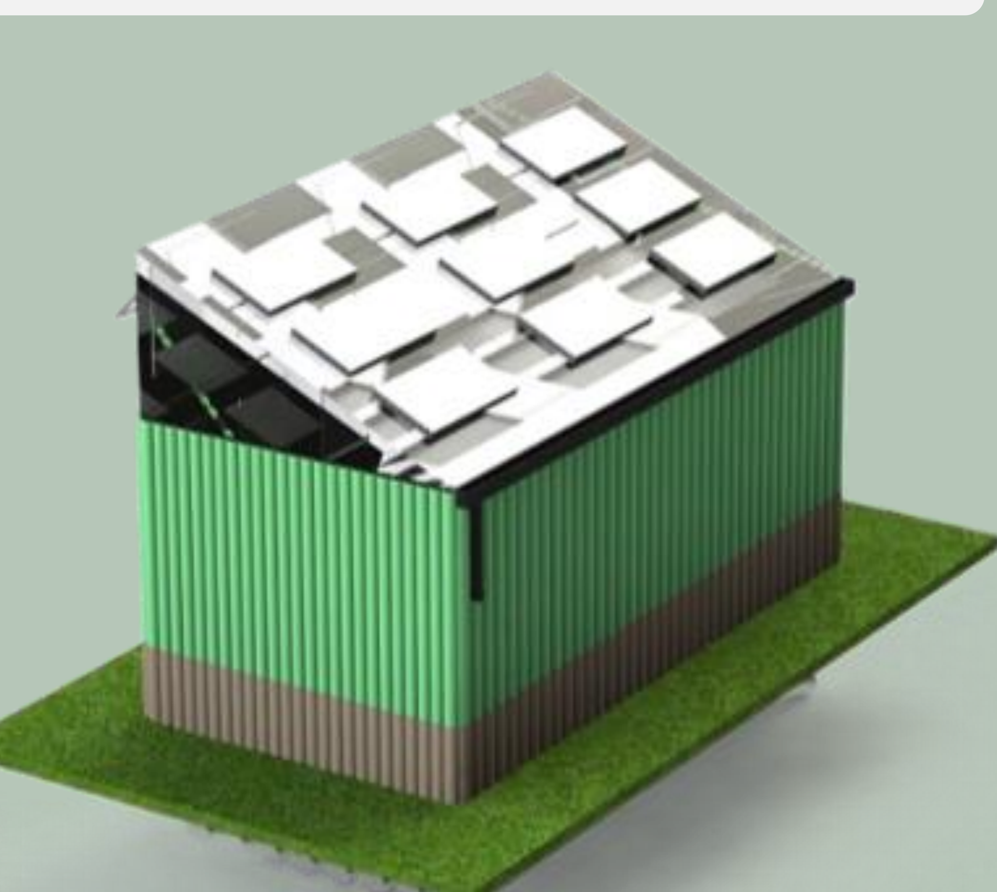
$$q_{Solar} = 600 \times 0.8 = 480\,\frac{W}{m^2}$$

Total solar gain through the walls:

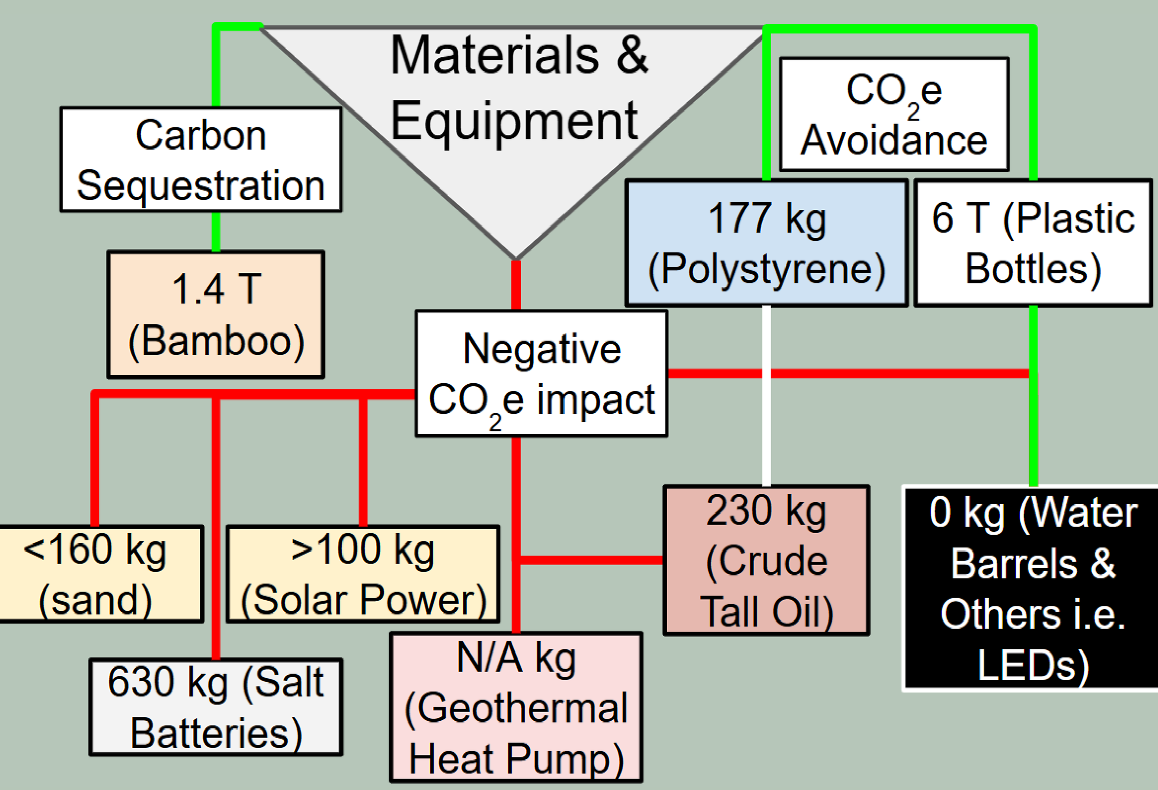
$$Q_{Solar} = 480 \times 1050 = 504\ kW$$

Final Design





CO_2e Impact



Total CO_2e Impact
Approximately negative 6,457 kg
PET wall, 41 Tonnes

Addressing The Scopes
1- More fuel-efficient machines for construction and production lowering CO_2e
2- Lowered by using solar power and GHP, creating renewables and decreasing energy use.
3- Will be addressed for each scenario due to transportation varying.

GHG Emissions = Activity Data × Emission Factor × Global Warming Potential (GWP)

Additional Information: Data used in calculations
CO_2e avoidance- Incineration of plastic, 2.9kg of CO2e/kg. (Global Alliance for Incinerator Alternatives, 2021)
Bamboo- 1cm diameter bamboo for each column, CO2e per m³ is -14.89kg, if accounting for embodied carbon (Zhao et al., 2019)
Polystyrene (PS)- Saved from incineration for making adhesive (Fraunhofer UMSICHT, 2016; Hearon et al., 2013)
Crude Tall Oil (CTO)- CO2e per kg is 0.94kg (Fraunhofer UMSICHT, 2016)
CTO&PS- Mixed at a 4:1 ratio.
Plastic Bottles- Using a 0.0682kg plastic bottle, 31.5cm high and 11cm wide (GT Online Shop, n.d.; Dimensions.com, n.d.), 30,560 are required.
Sand- Up to 0.2 kg of CO_2e/kg (Royal Commission, 2021)
Salt Battery- 62.68 kg of CO2e per kWh (Anton/Bauer, n.d.)
Solar Power- 19.13 grams of CO2e per watt required, calculated from the top two used materials.
Silicon, 2.4g per watt, 6kg CO2e/kg (Airis Energy, n.d.: Solaris Renewables, n.d.)
Copper, 1.05g per watt, 4.5kg CO2e/kg (Freeing Energy, n.d.; Thorlux Lighting, n.d.)
Geothermal Heat Pump (GHP)- Powered by renewable solar so less impacting than traditional heat regulation systems. (Kensa Heat Pumps, n.d.)
Water Barrels & Others- Smaller equipment such as LEDs would be countered by the CO_2e avoidance of reusing large plastic containers.
PET wall- 5mm thick requires 5.25 cubic meters equating to 6.982.5 kg and with a GWP of 5.91kg /kg (Sadeghi, Jeon and Seo, 2023: (CarbonCloud, n.d.)

Conclusion

In summary,
- Our sustainable greenhouse design supports key global sustainability goals by combining smart energy solutions and eco-friendly materials.
- A geothermal heat pump system is used to draw heat from the ground, helping to maintain comfortable temperatures inside throughout the year.
- This system reduces electricity use and reliance on fossil fuels, supporting **Goal 7: Affordable and Clean Energy**.
- Recycled plastic bottles filled with sand are used in the structure, offering low-carbon material benefits while improving insulation and strength, supporting **Goal 12: Responsible Consumption and Production**.
- The greenhouse reduces energy usage and carbon emissions, addressing **Goal 13: Climate Action**.
- With a stable indoor climate, it enables year-round food production using less energy, contributing to **Goal 2: Zero Hunger**.
- Overall, the greenhouse provides an efficient, sustainable way to grow food while helping protect the environment.

References

Anton/Bauer. (n.d.) 'Salt-E Dog - 9kWh Sustainable Sodium Power'. Available at: https://www.antonbauer./en/products/salt-e-dog/ (Accessed: 17 March 2025).
Airis Energy. (n.d.) 'Solar Panel Dimensions and Sizes: Complete Guide'. Available at: https://airisenergy.us/solar-panels-dimensions/ (Accessed: 17 March 2025).
Dimensions.com. (n.d.) 'Soda Bottle - 2 Liter Dimensions & Drawings'. Available at: https://www.dimensions.com/element/soda-bottle-2-liter (Accessed: 17 March 2025).
Fraunhofer UMSICHT. (2016) Analysis of the European Crude Tall Oil Industry – Environmental Impact, Socio-Economic Value & Downstream Potential. Available at: https://www.harrpa.eu/images/Publications/EU_CTO_Added_Value_Study_Fin.pdf (Accessed: 17 March 2025).
Freeing Energy. (n.d.) 'How much copper is used in US solar panels each year?'. Available at: https://www.freeingenergy.com/math/solar-panel-module-copper-weight-ton-us-global-m142/ (Accessed: 17 March 2025).
Global Alliance for Incinerator Alternatives (GAIA). (2021) Plastic is Carbon: Examining the Carbon Impacts of Plastics Throughout Their Lifecycle. Available at: https://www.no-burn.org/wp-content/uploads/Plastic-is-Carbon-Oct2021.pdf (Accessed: 17 March 2025).
GT Online Shop. (n.d.) '2 Litre Milk Bottle HDPE'. Available at: https://gtonline-shop.com/product/2-litre-milk-bottle-hdpe/ (Accessed: 17 March 2025).
Hearon, K., Nash, L.D., Rodriguez, J.N., Lonnecker, A.T., Raymond, J.E., Wilson, T.S., Wooley, K.L., and Maitland, D.J. (2013) 'A High-Performance Recycling Solution for Polystyrene Achieved by the Synthesis of Renewable Poly(thioether) Networks Derived from d-Limonene', Advanced Materials, 26(10), pp. 1552–1558. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC4000729/ (Accessed: 17 March 2025).
Kensa Heat Pumps. (n.d.) 'Gas Boilers Vs. Ground Source Heat Pumps'. Available at: https://kensa.co.uk/ground-source-heat-pumps/heat-pump-vs-gas-boiler (Accessed: 17 March 2025).
Solaris Renewables. (n.d.) 'What Is the Carbon Footprint of Solar Panel Manufacturing?'. Available at: https://solarisrenewables.com/blog/what-is-the-carbon-footprint-of-solar-panel-manufacturing/ (Accessed: 17 March 2025).
Thorlux Lighting. (n.d.) 'Switch to 'green' copper cuts carbon footprint'. Available at: https://www.thorlux.co.uk/news/green-copper (Accessed: 17 March 2025).
Zhao, X., Wu, J., Liang, C., and Li, C. (2019) 'Carbon Footprint Analysis of Bamboo Scrimber Flooring—Implications for Carbon Sequestration of Bamboo Forests and Its Products', Forests, 10(1), p. 51. Available at: https://www.mdpi.com/1999-4907/10/1/51 (Accessed: 17 March 2025).
Environment Agency. (2021). *State of the Environment: Soil Contamination*. Retrieved from https://www.gov.uk/environment-agency
GOV.UK (2019). Research and Statistics. [online] GOV.UK. Available at: https://www.gov.uk/search/research-and-statistics.
Royal Commission for the Exhibition of 1851 (n.d.) *Case Study: Finite*. Available at: https://royalcommission1851.org/case-study/finite (Accessed: 24 March 2025).
Heatable (2025) 'Solar Panel Costs UK (March 2025)', Heatable, 11 March. Available at: https://heatable.co.uk/solar/advice/solar-panel-costs (Accessed: 27 March 2025).
Greenfingers (n.d.) 'Horticultural Sand'. Available at: https://greenfingers.uk.com/product/horticultural-sand/ (Accessed: 27 March 2025).
IMS Heat Pumps (2024) 'What is the cost of installing a ground source heat pump in the UK?', IMS Heat Pumps, 28 March. Available at: https://www.imsheatpumps.co.uk/blog/cost-of-installing-ground-source-heat-pump-in-uk/ (Accessed: 27 March 2025).
UK Bamboo (n.d.) 'Bamboo Poles'. Available at: https://www.ukbamboo.com/shop/product-category/bamboo-poles/ (Accessed: 27 March 2025).
LED Supply and Fit (n.d.) '100m Strip Reel Dimmable LED Strip Lights | IP67'. Available at: https://ledsupplyandfit.co.uk/p/uk-best-100m-strip-reel-dimmable-led-strip-lights-ip67/ (Accessed: 27 March 2025).
Business Analytiq (n.d.) 'Crude Tall Oil Price Index'. Available at: https://businessanalytiq.com/procurementanalytics/index/tall-oil-price-index/ (Accessed: 27 March 2025).
McInerney, B. (2024) 'Salt Water Battery: Everything You Need To Know', Go Solar Quotes, 23 July. Available at: https://gosolarquotes.com.au/solar-battery-storage/salt/ (Accessed: 27 March 2025).
MET Office (2019). Weather and climate change - Met Office. [online] Met Office. Available at: https://www.metoffice.gov.uk/.
United Nations. (2023). *Sustainable Development Goals*. Retrieved from (https://www.un.org/sustainabledevelopment)
CarbonCloud (n.d.) *Product carbon footprint report: Tesco Still Water 2L PET bottle*. Available at: https://apps.carboncloud.com/climatehub/product-reports/id/3002489778300 (Accessed: 24 March 2025).
Sadeghi, K., Jeon, Y. and Seo, J. (2023) 'Roadmap to the sustainable synthesis of polymers: From the perspective of CO_2 upcycling', *Progress in Materials Science*, 135, 101103. Available at: https://doi.org/10.1016/j.pmatsci.2023.101103 (Accessed: 24 March 2025).

Royal Commission for the Exhibition of 1851 (n.d.) *Case Study: Finite*. Available at: https://royalcommission1851.org/case-study/finite (Accessed: 24 March 2025).
Heatable (2025) 'Solar Panel Costs UK (March 2025)', Heatable, 11 March. Available at: https://heatable.co.uk/solar/advice/solar-panel-costs (Accessed: 27 March 2025).
Greenfingers (n.d.) 'Horticultural Sand'. Available at: https://greenfingers.uk.com/product/horticultural-sand/ (Accessed: 27 March 2025).
IMS Heat Pumps (2024) 'What is the cost of installing a ground source heat pump in the UK?', IMS Heat Pumps, 28 March. Available at: https://www.imsheatpumps.co.uk/blog/cost-of-installing-ground-source-heat-pump-in-uk/ (Accessed: 27 March 2025).
UK Bamboo (n.d.) 'Bamboo Poles'. Available at: https://www.ukbamboo.com/shop/product-category/bamboo-poles/ (Accessed: 27 March 2025).
LED Supply and Fit (n.d.) '100m Strip Reel Dimmable LED Strip Lights | IP67'. Available at: https://ledsupplyandfit.co.uk/p/uk-best-100m-strip-reel-dimmable-led-strip-lights-ip67/ (Accessed: 27 March 2025).
Business Analytiq (n.d.) 'Crude Tall Oil Price Index'. Available at: https://businessanalytiq.com/procurementanalytics/index/tall-oil-price-index/ (Accessed: 27 March 2025).
McInerney, B. (2024) 'Salt Water Battery: Everything You Need To Know', Go Solar Quotes, 23 July. Available at: https://gosolarquotes.com.au/solar-battery-storage/salt/ (Accessed: 27 March 2025).

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